L07: Recap

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Supervised Learning

- Training, Validation and Testing
- Performance Metrics
- Problems:
  - Classification
  - Regression
- Algorithms
  - K-NN
  - Linear Classifier
  - Binary and Multi Class Classification
  - Probabilistic Classification (Bayesian)

Loss and Optimization

- Convex and Non-Convex Optimization
- Loss Functions
- Regularization
- Solutions:
  - Closed Form Solutions
  - Gradient Descent Solutions
  - Solutions that are Eigen Vectors

Regression

- Linear Regression
- Ridge Regression
- Lasso Regression

Rank and Factorization

- Data Matrtix and Low Rank
- Rank and Eigen Values
- SVD and Eigen Decomposition
- Low-Rank Approximation

Eigen Decomposition

Popular eigen expansion

$$A = \sum_{i=1}^{d} \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

Also

$$A = U \Lambda U^T$$

is known as the eigen decomposition.

SVD

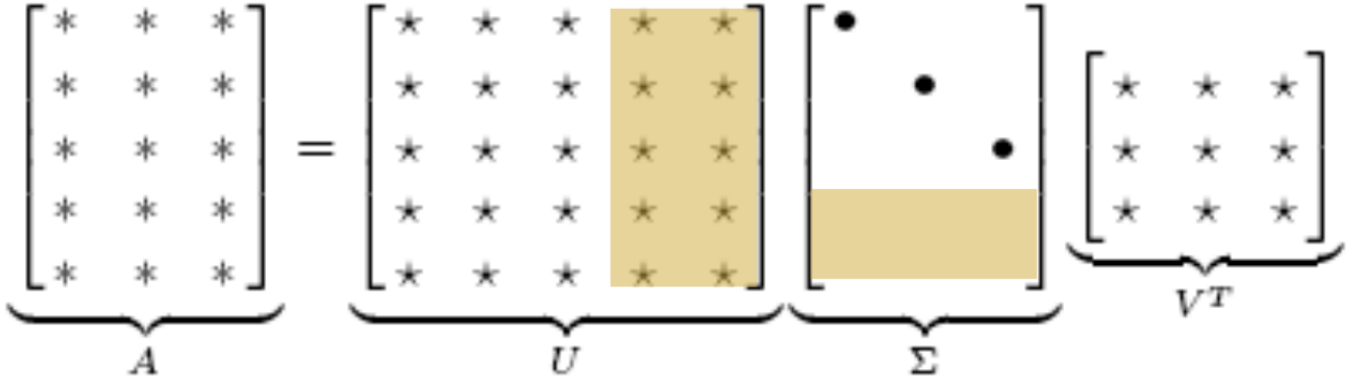
Formally, the **singular value decomposition** (SVD) of an $m \times n$ real or complex matrix M is a factorization of the form

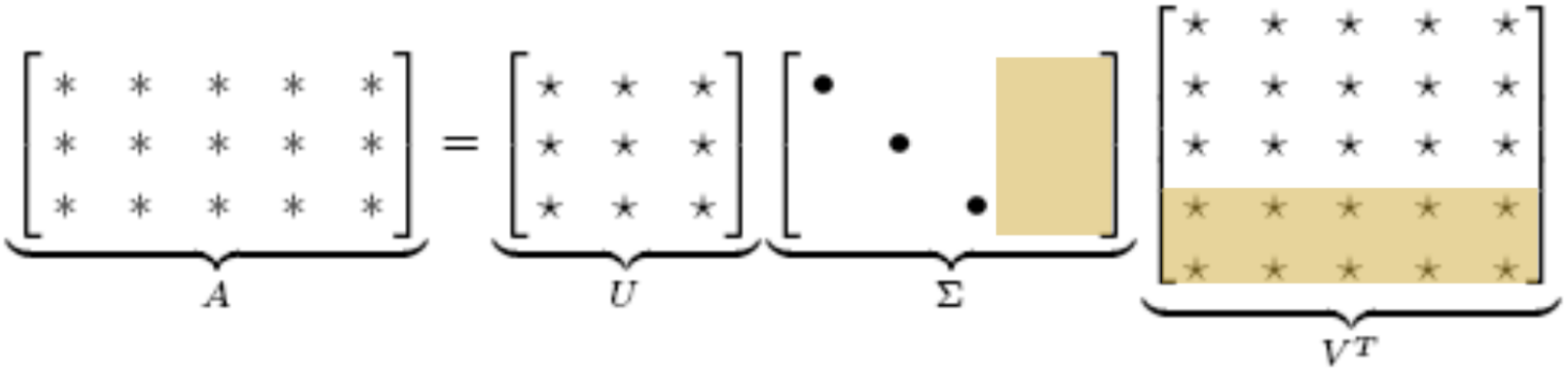
$$M = UDV^T$$

- U is $m \times n$, $U^T U = I_n$,
- D is diagonal $n \times n$ and
- V is $V^T V = VV^T = I_n$.

$$M = \sum_i D_i u_i v_i^T$$

SVD





PCA: Algorithm

1. Input: N samples of D dimension
2. Compute Covariance Matrix $$\Sigma = \frac{1}{N}\sum_{i=1}^{N}[x_i - \mu][x_i - \mu]^T$$
3. Compute Eigen Values and Eigen Vectors of Covariance Matrix
4. Select `d' eigen vectors corresponding to 'd' largest eigen values
5. Arrange them as rows of 'A' (a 'd' X 'D' matrix)
6. Find lower dimensional representations as: $x' = Ax$

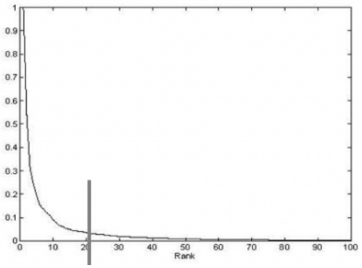
How many Eigen Values?

How many eigen values are required so that "most" of the information in the covariance/data is preserved? Consider the popular eigen expansion:

$$\Sigma = \sum_{i=1}^{D} \lambda_i v_i v_i^T$$
$$= \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T + \cdots + \lambda_{N-1} v_{N-1} v_{N-1}^T + \lambda_N v_N v_N^T$$

Eigen values (Lambda) are in decreasing order. Even if we discard the small eigen values, most information is preserved.

Typical Eigen Value Spectrum



$$d = \min_p \frac{\sum_{i=1}^{p} \lambda_i}{\sum_{i=1}^{D} \lambda_i} \geq 0.95$$

Problem

Consider face images of size 100×100 and we have 200 such images. Assume means are subtracted.

1. What is the size of the covariance matrix?
2. What is the rank of the covariance matrix? (guess!)
3. What is the size of XX^T and X^TX and what are their ranks?
4. How are the Eigen values of X^TX and XX^T related?
5. How are the Eigen vectors of X^TX and XX^T related?

Quiz - 1

Quiz 1: Best Wishes!!